

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(c)	(i)	Any 3 of 4 Comment regarding half-life e.g. long enough to enable measurement / short so effect soon disappears (1) Gamma emitter so can be detected out of the body (1) Daughter product stable (1) Low ionisation (1)			3	3		
		(ii)	Gamma rays pass through a <u>collimator</u> (1) Cause scintillations / light flashes on a crystal (1) Collected by a series of photomultiplier tubes OR CCD (1)	3			3		
		(iii)	Mass-energy of electron [or positron] should be equal to the photon energy [because $2e \rightarrow 2\gamma$] (1) $E_{\text{electron}} = mc^2 = 8.20 \times 10^{-14} \text{ J}$ (1) $\therefore E_{\text{ph}} = \frac{8.20 \times 10^{-14} [\text{J}]}{1.6 \times 10^{-13} [\text{J MeV}^{-1}]} = 0.512 \text{ MeV}$		1	1	3	2	
			Question 7 total	6	9	5	20	10	0